Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Question Bank 6.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, can the three lengths make a triangle?

1. 3 m, 9 m, 11 m

Yes — 3 + 9 = 12 > 11

2. 5 inches, 13 inches, 7 inches

No — 5 + 7 = 12 < 13

The longest is listed second.

3. 8 ft, 4 ft, 4 ft

No — 4 + 4 = 8 lies flat

The equal case.

For questions 4 and 5, two sides of a triangle are given. Write an inequality for the third side, x.

4. 3 cm and 8 cm

5 < x < 11 (centimeters)

5. 1 cm and 5 cm; the third side is a whole number of centimeters — which one?

4 < x < 6, so x = 5 cm — the only whole number between

For questions 6 through 8, do the lengths make a right triangle?

6. 4.5 cm, 6 cm, 7.5 cm

Yes — $20.25 + 36 = 56.25 = 7.5^2$

7. 26 inches, 24 inches, 10 inches

Yes — $10^2 + 24^2 = 676 = 26^2$

The longest is listed first.

8. 7 m, 10 m, 12 m

No — 49 + 100 = 149, not 144

Choose one.

9. Which set of lengths makes a right triangle?

- ○ A. 4, 5, 6
- ○ B. 2, 3, 5
- ○ C. 6, 8, 12
- ● **D. 11, 60, 61**

D

$121 + 3{,}600 = 3{,}721 = 61^2$.